

## Jawless fish

Class *Cyclostomata*

### Sea lamprey

*Petromyzon marinus*

- ↔ Up to 4 ft (1.2 m)
- 📍 North Atlantic, Mediterranean
- ⊗ Locally common

This is the largest species of lamprey. An adult sea lamprey uses its suckerlike mouth—armed with horny teeth—to clamp onto the sides of another fish, scrape a hole in the host's skin, and suck out flesh and fluids. Sexually mature individuals migrate upriver to spawn in fresh water. Here, larvae may live for more than five years before maturing into adults and moving out to sea. Lampreys breed only once and die after spawning.

## Cartilaginous fish

Class *Chondrichthyes*

### Spotted wobbegong

*Orectolobus maculatus*

- ↔ 4–6 ft (1.2–1.7 m)
- 📍 S. Australia
- ⊗ Near threatened

Flattened like a ray, wobbegongs—sluggish inhabitants of tropical inshore waters of the Indo-Pacific—are among the most distinctive of sharks. This species' mottled skin pattern and tasseled body help it to blend in with seaweed and coral—ideally suited for ambushing smaller fish. It hunts on the bottom, sometimes using its paired pectoral fins to climb between rock pools, with parts of its body showing above the water.

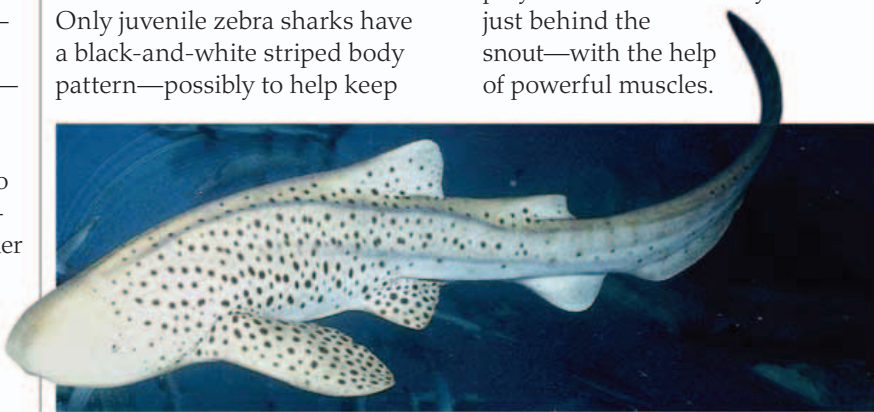
### Zebra shark

*Stegostoma fasciatum*

- ↔ 5–8 ft (1.5–2.4 m)
- 📍 Indo-Pacific
- ⊗ Vulnerable

Only juvenile zebra sharks have a black-and-white striped body pattern—possibly to help keep

predators away by mimicking the highly venomous banded sea snake. In adults, the stripes are replaced by leopardlike spots. A suction feeder, the shark draws bottom-living fish and invertebrate prey into its mouth cavity—located just behind the snout—with the help of powerful muscles.



### Blacktip reef shark

*Carcharhinus melanopterus*

- ↔ Up to 3–4 ft (0.9–1.2 m)
- 📍 Tropical Indo-Pacific, E. Mediterranean
- ⊗ Near threatened

Named for its black-tipped fins—its dorsal fin is often seen projecting above the surface—this is one of the most common predatory sharks of tropical Asian and Australasian waters. Individuals spend much of their time in a territorial patch of reef. Pregnant females move into shallow, sand-bottomed lagoons to give birth.

### Great white shark

*Carcharodon carcharias*

- ↔ 12–17 ft (3.7–5.2 m)
- 📍 Temperate and tropical waters worldwide; at times in cold waters
- ⊗ Vulnerable



A formidable hunter, the great white shark has one of the most powerful bites of any living animal. It is a particularly efficient predator of marine mammals, such as seals, sea lions, and dolphins, although it will feed on any large creature that it can catch. A core of heat-generating body muscles keep it alert and active even in cold waters, while circulatory modifications help retain the heat. Although usually solitary, it may sometimes be seen in pairs or small groups feeding at a carcass, with larger individuals eating first.

### Leopard shark

*Triakis semifasciata*

- ↔ Up to 7 ft (2.1 m)
- 📍 E. North Pacific
- ⊗ Locally common

The spots on the leopard shark's body may help camouflage it among the weeds and corals of coastal waters. It hunts in schools for small fish and invertebrates, sometimes venturing into estuaries and bays as well. The shark shuns colder waters, and northern populations migrate to greater depths in winter, where temperatures are less extreme.

### Smooth hammerhead

*Sphyrna zygaena*

- ↔ 7–8 ft (2.1–2.4 m)
- 📍 Tropical, subtropical, and temperate waters worldwide
- ⊗ Vulnerable

The anatomy of a large hammerhead shark's skull is likely to provide extra lift as the shark swims. The "blade" of the head helps with maneuverability, while a battery of chemical and electrical receptors along its front edge detects prey, especially bottom-living stingrays.

### Giant manta ray

*Manta birostris*

- ↔ 13–23 ft (4–7 m)
- 📍 Tropical and sometimes warm temperate waters worldwide
- ⊗ Vulnerable

The world's biggest ray virtually "flies" underwater by flapping its giant pectoral fins. The manta has vestigial teeth that are useless for processing food. Instead, it is a filter feeder—with the help of two lobe-like horns, small planktonic animals are channeled into the mouth and then trapped by the gill rakers.

